Lesson 19: Inner Product Spaces and Hilbert Spaces

AI/ML Mathematics

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1 Motivation

Normed spaces provide a notion of size (the norm), but they lack geometry. An inner product adds the crucial concepts of angle and orthogonality, enabling projections, decompositions, and least-squares approximation. A Hilbert space—a complete inner product space—is the richest and most well-behaved infinite-dimensional space, enjoying a geometry that directly generalizes finite-dimensional Euclidean space.

The passage from normed spaces to inner product spaces is not merely an abstraction. In reinforcement learning and machine learning, inner product structure arises naturally:

- **Least-squares value function approximation:** Projecting the Bellman target onto a linear subspace requires the inner product on L^2 (or $\mathbb{R}^n$ with a weighted inner product).
- **Reproducing kernel Hilbert spaces (RKHS):** Kernel methods operate in Hilbert spaces of functions, where the representer theorem guarantees finite-dimensional solutions.
- **Natural gradient methods:** The Fisher information matrix defines an inner product on the parameter space, and the natural gradient is the steepest descent direction in this geometry.

Why Inner Products Matter for RL/ML

1. **LSTD (Least-Squares Temporal Difference):** The fixed-point equation $\Phi w = \Pi_\mu T(\Phi w)$ involves the orthogonal projection Π_μ onto the column space of Φ in the $L^2(\mu)$-inner product. Understanding this projection requires the Hilbert space structure of $L^2(\mu)$.
2. **Kernel methods:** A positive-definite kernel $k(x, y)$ defines a unique RKHS $\mathcal{H}_k$. The reproducing property $f(x) = \langle f, k(\cdot, x)\rangle_{\mathcal{H}_k}$ is an application of the Riesz Representation Theorem on Hilbert spaces.
3. **Gradient geometry:** The Cauchy–Schwarz inequality, proved in this lesson, is the foundation of gradient-based optimization: it identifies the direction of steepest ascent of a linear functional on a Hilbert space.

Prerequisites. Metric spaces, normed spaces, Banach spaces, and compactness (Lessons 15–18).

Outline. Section 2 defines inner product spaces and establishes their fundamental properties. Section 3 proves the Cauchy–Schwarz inequality with a complete proof. Section 4 develops the parallelogram law and polarization identity. Section 5 introduces Hilbert spaces and key examples. Section 6 develops orthogonality and orthogonal complements. Section 7 provides exercises.

2 Inner Product Spaces

Definition 2.1 (Inner Product)**.** Let X be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An *inner product* on X is a map $\langle \cdot, \cdot \rangle \colon X \times X \to \mathbb{F}$ satisfying, for all $x, y, z \in X$ and all $\alpha, \beta \in \mathbb{F}$:

(IP1) **Linearity in the first argument:** $\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle$.

(IP2) **Conjugate symmetry:** $\langle x, y\rangle = \overline{\langle y, x\rangle}$.

(IP3) **Positive definiteness:** $\langle x, x\rangle \geq 0$, with equality if and only if $x = 0$.

A vector space equipped with an inner product is called an *inner product space* (or *pre-Hilbert space*). See Kreyszig [1989] for an extensive treatment.

Remark 2.2 (Consequences of the Axioms). From (IP1) and (IP2) we derive conjugate-linearity in the second argument:

$$\langle x, \alpha y + \beta z\rangle = \overline{\langle \alpha y + \beta z, x\rangle} = \overline{\alpha\langle y, x\rangle + \beta\langle z, x\rangle} = \bar{\alpha}\langle x, y\rangle + \bar{\beta}\langle x, z\rangle.$$

In the real case ($\mathbb{F} = \mathbb{R}$), conjugate symmetry reduces to symmetry $\langle x, y\rangle = \langle y, x\rangle$, and the inner product is bilinear.

Remark 2.3 (Convention). We adopt the convention that the inner product is linear in the first argument and conjugate-linear in the second (the "physics convention"). Some texts, particularly in physics, use the opposite convention. The choice does not affect the theory; one must simply be consistent.

Definition 2.4 (Induced Norm)**.** Given an inner product space $(X, \langle\cdot,\cdot\rangle)$, the *induced norm* is defined by

$$\|x\| = \langle x, x\rangle^{1/2}.$$

We verify below (after proving Cauchy–Schwarz) that this is indeed a norm.

2.1 Standard Examples

Example 2.5 (Euclidean Space $\mathbb{R}^n$)**.** The standard inner product on $\mathbb{R}^n$ is

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i y_i = x^\top y.$$

The induced norm is the Euclidean norm $\|x\| = \left(\sum_{i=1}^n x_i^2\right)^{1/2}$. Verification of (IP1)–(IP3) is immediate.

Example 2.6 (Weighted Inner Product on $\mathbb{R}^n$)**.** Let $D = \operatorname{diag}(d_1, \ldots, d_n)$ with all $d_i > 0$. Then

$$\langle x, y\rangle_D = x^\top D\, y = \sum_{i=1}^{n} d_i\, x_i\, y_i$$

is an inner product on $\mathbb{R}^n$. The induced norm is $\|x\|_D = \left(\sum_{i=1}^n d_i\, x_i^2\right)^{1/2}$.

Weighted Inner Products in RL

In temporal-difference learning with linear function approximation, the projection Π_μ onto the feature subspace uses the weighted inner product $\langle f, g\rangle_\mu = \sum_{s\in\mathcal{S}} \mu(s)\, f(s)\, g(s)$, where μ is the stationary distribution of the policy. This is precisely the inner product of Example 2.6 with $d_s = \mu(s)$.

Example 2.7 (The Sequence Space ℓ^2)**.** The space ℓ^2 consists of all sequences $x = (x_1, x_2, \ldots)$ of real (or complex) numbers with $\sum_{i=1}^\infty |x_i|^2 < \infty$. The inner product is

$$\langle x, y\rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}.$$

Well-definedness: By the Cauchy–Schwarz inequality for finite sums (applied to the partial sums and then taking limits), the series converges absolutely: $\sum_{i=1}^\infty |x_i\overline{y_i}| = \sum_{i=1}^\infty |x_i||y_i| \le \left(\sum |x_i|^2\right)^{1/2}\left(\sum |y_i|^2\right)^{1/2} < \infty$. The induced norm is $\|x\| = \left(\sum_{i=1}^\infty |x_i|^2\right)^{1/2}$.

Example 2.8 (The Function Space $L^2(\mu)$)**.** Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. The space $L^2(\mu)$ consists of (equivalence classes of) measurable functions $f\colon \Omega \to \mathbb{F}$ with $\int |f|^2\, d\mu < \infty$. The inner product is

$$\langle f, g\rangle = \int_\Omega f\, \overline{g}\, d\mu.$$

The induced norm is $\|f\|_2 = \left(\int |f|^2\, d\mu\right)^{1/2}$.

Example 2.9 (Matrix Inner Product (Frobenius))**.** On $\mathbb{R}^{m\times n}$, the Frobenius inner product is

$$\langle A, B\rangle_F = \operatorname{tr}(B^\top A) = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}B_{ij}.$$

The induced norm is the Frobenius norm $\|A\|_F = \left(\sum_{i,j} A_{ij}^2\right)^{1/2}$.

Not Every Norm Comes from an Inner Product

The spaces ℓ^p and L^p for $p \neq 2$ are normed spaces but *not* inner product spaces. We will see (Theorem 4.1) that a norm arises from an inner product if and only if it satisfies the parallelogram law, which fails for $p \neq 2$.

3 The Cauchy–Schwarz Inequality

The Cauchy–Schwarz inequality is arguably the single most important inequality in analysis. It appears in every branch of mathematics and its applications to ML are ubiquitous: from bounding inner products in gradient computations to establishing the well-definedness of cosine similarity.

Theorem 3.1 (Cauchy–Schwarz Inequality)**.** *Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For all $x, y \in X$,*

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\|,$$

where $\|x\| = \langle x, x\rangle^{1/2}$. Equality holds if and only if x and y are linearly dependent (i.e., one is a scalar multiple of the other, or one of them is zero).

Proof. If $y = 0$, both sides are zero and the inequality holds trivially (and equality holds with $x = x \cdot 0 + 0 \cdot y$ trivially dependent when one vector is zero).

Assume $y \neq 0$. For any scalar $\alpha \in \mathbb{F}$, positive definiteness gives

$$0 \leq \langle x - \alpha y, x - \alpha y\rangle = \|x\|^2 - \alpha\langle y, x\rangle - \bar{\alpha}\langle x, y\rangle + |\alpha|^2\|y\|^2. \tag{1}$$

We choose α to minimize the right-hand side. The optimal choice is

$$\alpha = \frac{\langle x, y\rangle}{\|y\|^2}. \tag{2}$$

Then $\bar{\alpha} = \overline{\langle x, y\rangle}/\|y\|^2 = \langle y, x\rangle/\|y\|^2$ (by conjugate symmetry), and $|\alpha|^2 = |\langle x, y\rangle|^2/\|y\|^4$. Substituting into (1):

$$\begin{aligned}
0 &\leq \|x\|^2 - \frac{\langle x, y\rangle}{\|y\|^2}\langle y, x\rangle - \frac{\langle y, x\rangle}{\|y\|^2}\langle x, y\rangle + \frac{|\langle x, y\rangle|^2}{\|y\|^4}\|y\|^2 \\
&= \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2} - \frac{|\langle x, y\rangle|^2}{\|y\|^2} + \frac{|\langle x, y\rangle|^2}{\|y\|^2} \\
&= \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}.
\end{aligned}$$

Rearranging gives $|\langle x, y\rangle|^2 \leq \|x\|^2\|y\|^2$. Taking square roots yields

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\|.$$

Equality condition. If x and y are linearly dependent, say $x = \beta y$, then $|\langle x, y\rangle| = |\beta|\|y\|^2 = \|x\|\|y\|$.

Conversely, if equality holds and $y \neq 0$, then $\langle x - \alpha y, x - \alpha y\rangle = 0$ with α as in (2). By positive definiteness, $x - \alpha y = 0$, i.e., $x = \alpha y$. □

The Cauchy–Schwarz Inequality

For all x, y in an inner product space: $|\langle x, y\rangle| \leq \|x\| \cdot \|y\|$, with equality if and only if x and y are linearly dependent. This single inequality implies:

- the triangle inequality for the induced norm,
- the well-definedness of cosine similarity,
- the identification of the gradient direction as steepest ascent.

Corollary 3.2 (Triangle Inequality for the Induced Norm)**.** *For all x, y in an inner product space, $\|x + y\| \leq \|x\| + \|y\|$.*

Proof. We compute:

$$\begin{aligned}
\|x+y\|^2 &= \langle x+y, x+y\rangle \\
&= \|x\|^2 + \langle x, y\rangle + \langle y, x\rangle + \|y\|^2 \\
&= \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \\
&\leq \|x\|^2 + 2|\langle x, y\rangle| + \|y\|^2 \\
&\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \quad \text{(Cauchy–Schwarz)} \\
&= (\|x\| + \|y\|)^2.
\end{aligned}$$

Taking square roots gives the result. □

Remark 3.3. The induced norm $\|x\| = \langle x, x\rangle^{1/2}$ is indeed a norm:

- *Positive definiteness:* $\|x\| = 0$ iff $\langle x, x\rangle = 0$ iff $x = 0$ (by IP3).
- *Homogeneity:* $\|\alpha x\| = \langle \alpha x, \alpha x\rangle^{1/2} = (|\alpha|^2\|x\|^2)^{1/2} = |\alpha|\|x\|$.
- *Triangle inequality:* Corollary 3.2.

Thus every inner product space is automatically a normed space (and hence a metric space with $d(x, y) = \|x - y\|$).

4 Parallelogram Law and Polarization Identity

The parallelogram law characterizes precisely which norms come from inner products. The polarization identity provides the formula for recovering the inner product from the norm.

Theorem 4.1 (Parallelogram Law)**.** *In any inner product space,*

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2) \quad \textit{for all } x, y \in X.$$

Proof. We expand each term using the inner product:

$$\begin{aligned}
\|x+y\|^2 &= \langle x+y, x+y\rangle = \|x\|^2 + \langle x, y\rangle + \langle y, x\rangle + \|y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2, \\
\|x-y\|^2 &= \langle x-y, x-y\rangle = \|x\|^2 - \langle x, y\rangle - \langle y, x\rangle + \|y\|^2 = \|x\|^2 - 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2.
\end{aligned}$$

Adding these two equations:

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$ □

Remark 4.2 (Geometric Interpretation). The parallelogram law states that the sum of the squares of the diagonals of a parallelogram equals the sum of the squares of its four sides. This is the defining geometric property of inner product spaces.

Theorem 4.3 (Jordan–von Neumann Theorem (Converse))**.** *If a normed space $(X, \|\cdot\|)$ satisfies the parallelogram law, then the norm is induced by an inner product, which is recovered by the polarization identity.*

The proof of this theorem is somewhat technical (one must verify all the inner product axioms for the candidate recovered by polarization) and is omitted. We state and prove the polarization identity itself.

Proposition 4.4 (Polarization Identity)**.** *In a real inner product space,*

$$\langle x, y\rangle = \frac{1}{4}\big(\|x+y\|^2 - \|x-y\|^2\big). \tag{3}$$

In a complex inner product space,

$$\langle x, y\rangle = \frac{1}{4}\sum_{k=0}^{3} i^k \|x + i^k y\|^2. \tag{4}$$

Proof. Real case. Using the expansion from the parallelogram law proof:

$$\|x+y\|^2 - \|x-y\|^2 = (\|x\|^2 + 2\langle x, y\rangle + \|y\|^2) - (\|x\|^2 - 2\langle x, y\rangle + \|y\|^2) = 4\langle x, y\rangle.$$

Dividing by 4 gives (3).

Complex case. Compute $\sum_{k=0}^{3} i^k \|x + i^k y\|^2$. For each k:

$$\|x + i^k y\|^2 = \|x\|^2 + i^k\langle y, x\rangle + \overline{i^k}\langle x, y\rangle + |i^k|^2\|y\|^2.$$

Since $|i^k| = 1$ and $\overline{i^k} = i^{-k} = (-i)^k$, we have $\overline{i^k} = \bar{i}^k = (-i)^k$. Multiplying by i^k and summing:

$$\sum_{k=0}^{3} i^k\|x + i^k y\|^2 = \Big(\sum_{k=0}^{3} i^k\Big)\|x\|^2 + \Big(\sum_{k=0}^{3} i^{2k}\Big)\langle y, x\rangle + \Big(\sum_{k=0}^{3} i^k(-i)^k\Big)\langle x, y\rangle + \Big(\sum_{k=0}^{3} i^k\Big)\|y\|^2.$$

Now $\sum_{k=0}^{3} i^k = 1+i+i^2+i^3 = 1+i-1-i = 0$ and $\sum_{k=0}^{3} i^{2k} = 1+i^2+i^4+i^6 = 1-1+1-1 = 0$. Also, $i^k(-i)^k = (i\cdot(-i))^k = 1^k = 1$, so $\sum_{k=0}^{3} i^k(-i)^k = 4$. Therefore the entire sum equals $4\langle x, y\rangle$. □

Polarization in Optimization

The polarization identity shows that the inner product—and hence angles and orthogonality—can be recovered purely from norm computations. In practice, this means that if we can evaluate $\|v + w\|$ and $\|v - w\|$ for value function approximations, we can compute their inner product without direct access to the integral $\langle v, w\rangle = \int v\, w\, d\mu$.

5 Hilbert Spaces

Definition 5.1 (Hilbert Space)**.** A *Hilbert space* is an inner product space that is complete with respect to the norm induced by its inner product. Equivalently, a Hilbert space is a Banach space whose norm satisfies the parallelogram law.

Proposition 5.2 (Continuity of the Inner Product)**.** *The inner product is jointly continuous: if $x_n \to x$ and $y_n \to y$ in an inner product space, then $\langle x_n, y_n\rangle \to \langle x, y\rangle$.*

Proof. We estimate:

$$\begin{aligned}|\langle x_n, y_n\rangle - \langle x, y\rangle| &= |\langle x_n, y_n\rangle - \langle x, y_n\rangle + \langle x, y_n\rangle - \langle x, y\rangle| \\ &\le |\langle x_n - x, y_n\rangle| + |\langle x, y_n - y\rangle| \\ &\le \|x_n - x\|\cdot\|y_n\| + \|x\|\cdot\|y_n - y\| \quad \text{(Cauchy–Schwarz)}.\end{aligned}$$

Since $y_n \to y$, the sequence $(\|y_n\|)$ is bounded, say $\|y_n\| \le M$ for all n. Then

$$|\langle x_n, y_n\rangle - \langle x, y\rangle| \le M\|x_n - x\| + \|x\|\|y_n - y\| \to 0.$$ □

Remark 5.3. In fact, each "slot" of the inner product is separately continuous (and linear/conjugate-linear), and by Cauchy–Schwarz, each evaluation $x \mapsto \langle x, y\rangle$ is a bounded linear functional with $\|\langle\cdot, y\rangle\| = \|y\|$. This observation leads to the Riesz Representation Theorem (Lesson 20).

5.1 Standard Hilbert Spaces

Example 5.4 (Finite-Dimensional: $\mathbb{R}^n$ and $\mathbb{C}^n$)**.** $\mathbb{R}^n$ with the Euclidean inner product is a Hilbert space. More generally, $\mathbb{R}^n$ with any weighted inner product $\langle x, y\rangle_D = x^\top D\, y$ (with D positive definite) is a Hilbert space, since all norms on $\mathbb{R}^n$ are equivalent and $\mathbb{R}^n$ is complete.

Example 5.5 (The Sequence Space ℓ^2)**.** The space ℓ^2 with the inner product $\langle x, y\rangle = \sum_{i=1}^{\infty} x_i \bar{y}_i$ is a Hilbert space. Completeness: if $(x^{(n)})$ is Cauchy in ℓ^2, then for each fixed i, the coordinate sequence $(x_i^{(n)})_{n\geq 1}$ is Cauchy in $\mathbb{F}$ (since $|x_i^{(n)} - x_i^{(m)}| \leq \|x^{(n)} - x^{(m)}\|$), hence convergent to some x_i^*. One then verifies that $x^* = (x_i^*)_{i\geq 1} \in \ell^2$ and $\|x^{(n)} - x^*\| \to 0$.

Example 5.6 (The Function Space $L^2(\mu)$)**.** For any σ-finite measure space $(\Omega, \mathcal{A}, \mu)$, the space $L^2(\mu)$ with the inner product $\langle f, g\rangle = \int f\, \bar{g}\, d\mu$ is a Hilbert space. Completeness is the content of the Riesz–Fischer theorem: every Cauchy sequence in $L^2(\mu)$ converges in $L^2(\mu)$ (see Brézis [2011] for details).

The Three Fundamental Hilbert Spaces

The three most important Hilbert spaces for RL/ML are:

1. $\mathbb{R}^n$ (finite-state value functions, parameter spaces),
2. ℓ^2 (countable-state value functions, neural network weight spaces with ℓ^2 regularization),
3. $L^2(\mu)$ (continuous-state value functions, policy gradients with respect to a measure μ).

Every separable Hilbert space is isometrically isomorphic to ℓ^2 (or to $\mathbb{R}^n$ if finite-dimensional) Conway [1990]. This remarkable fact means that, up to isomorphism, there is essentially one infinite-dimensional separable Hilbert space.

$C[0,1]$ with the L^2 Inner Product is Not a Hilbert Space

The space $C[0,1]$ with the inner product $\langle f, g\rangle = \int_0^1 f(t)\, g(t)\, dt$ is an inner product space but *not* a Hilbert space. One can construct a Cauchy sequence of continuous functions whose L^2 limit is a discontinuous function (for instance, approximate the indicator function $\mathbf{1}_{[0,1/2]}$ by continuous functions). The completion of this space is $L^2([0,1])$.

6 Orthogonality and Orthogonal Complements

Orthogonality is the geometric heart of Hilbert space theory. It enables decompositions, projections, and the entire theory of least-squares approximation.

Definition 6.1 (Orthogonality)**.** Two elements x, y in an inner product space are *orthogonal*, written $x \perp y$, if $\langle x, y\rangle = 0$. A subset $S \subseteq X$ is an *orthogonal set* if any two distinct elements of S are orthogonal. An orthogonal set in which every element has norm 1 is an *orthonormal set*.

Theorem 6.2 (Pythagorean Theorem)**.** *If $x_1, \ldots, x_n$ are pairwise orthogonal elements of an inner product space, then*

$$\Big\|\sum_{k=1}^{n} x_k\Big\|^2 = \sum_{k=1}^{n} \|x_k\|^2.$$

Proof. Expand the left-hand side using the inner product:

$$\Big\|\sum_{k=1}^{n} x_k\Big\|^2 = \Big\langle \sum_{k=1}^{n} x_k, \sum_{j=1}^{n} x_j \Big\rangle = \sum_{k=1}^{n}\sum_{j=1}^{n} \langle x_k, x_j\rangle.$$

Since $\langle x_k, x_j\rangle = 0$ for $k \neq j$ (by orthogonality), only the diagonal terms survive:

$$\sum_{k=1}^{n}\sum_{j=1}^{n}\langle x_k, x_j\rangle = \sum_{k=1}^{n}\langle x_k, x_k\rangle = \sum_{k=1}^{n}\|x_k\|^2. \qquad \square$$

Remark 6.3 (Infinite Pythagorean Theorem). If $(x_k)_{k=1}^{\infty}$ is a sequence of pairwise orthogonal elements in a Hilbert space and $\sum_{k=1}^{\infty}\|x_k\|^2 < \infty$, then the series $\sum_{k=1}^{\infty} x_k$ converges (by completeness, since the partial sums form a Cauchy sequence by the Pythagorean theorem), and $\left\|\sum_{k=1}^{\infty} x_k\right\|^2 = \sum_{k=1}^{\infty}\|x_k\|^2$.

Definition 6.4 (Orthogonal Complement)**.** For a subset S of an inner product space X, the *orthogonal complement* of S is

$$S^{\perp} = \{x \in X : \langle x, s\rangle = 0 \text{ for all } s \in S\}.$$

Proposition 6.5 (Properties of Orthogonal Complements)**.** *Let S and T be subsets of an inner product space X.*

(i) *$S^{\perp}$ is a closed subspace of X.*
(ii) *If $S \subseteq T$, then $T^{\perp} \subseteq S^{\perp}$.*
(iii) *$S \cap S^{\perp} \subseteq \{0\}$.*
(iv) *$S \subseteq (S^{\perp})^{\perp}$.*
(v) *$S^{\perp} = (\overline{\operatorname{span}(S)})^{\perp}$.*

Proof. **(i)** *Subspace:* For $x_1, x_2 \in S^{\perp}$ and scalars α, β: $\langle \alpha x_1 + \beta x_2, s\rangle = \alpha\langle x_1, s\rangle + \beta\langle x_2, s\rangle = 0$ for all $s \in S$.

Closed: Let (x_n) be a sequence in $S^{\perp}$ with $x_n \to x$. For any $s \in S$: $\langle x, s\rangle = \lim_{n\to\infty}\langle x_n, s\rangle = \lim_{n\to\infty} 0 = 0$ (by continuity of the inner product, Proposition 5.2). So $x \in S^{\perp}$.

(ii) If $x \in T^{\perp}$, then $\langle x, t\rangle = 0$ for all $t \in T$. Since $S \subseteq T$, this holds for all $s \in S$, so $x \in S^{\perp}$.

(iii) If $x \in S \cap S^{\perp}$, then $\langle x, x\rangle = 0$ (since $x \in S$ and $x \in S^{\perp}$), so $x = 0$ by positive definiteness.

(iv) If $s \in S$ and $x \in S^{\perp}$, then $\langle s, x\rangle = \overline{\langle x, s\rangle} = 0$. Hence $s \in (S^{\perp})^{\perp}$.

(v) One inclusion: $S \subseteq \overline{\operatorname{span}(S)}$ implies $(\overline{\operatorname{span}(S)})^{\perp} \subseteq S^{\perp}$ by (ii).

Other inclusion: Let $x \in S^{\perp}$. Then $\langle x, s\rangle = 0$ for all $s \in S$. By linearity, $\langle x, y\rangle = 0$ for all $y \in \operatorname{span}(S)$. By continuity of the inner product, $\langle x, y\rangle = 0$ for all $y \in \overline{\operatorname{span}(S)}$. Hence $x \in (\overline{\operatorname{span}(S)})^{\perp}$. $\square$

Remark 6.6. In a Hilbert space, we have the stronger result $(S^{\perp})^{\perp} = \overline{\operatorname{span}(S)}$. This is a consequence of the Projection Theorem (proved in Lesson 20), which states that every closed subspace M of a Hilbert space satisfies $X = M \oplus M^{\perp}$. The inclusion $S \subseteq (S^{\perp})^{\perp}$ from (iv) above may be strict in incomplete inner product spaces.

Proposition 6.7 (Orthogonal Elements are Linearly Independent)**.** *If $\{x_1, \ldots, x_n\}$ is an orthogonal set of non-zero vectors, then $x_1, \ldots, x_n$ are linearly independent.*

Proof. Suppose $\sum_{k=1}^{n}\alpha_k x_k = 0$. Taking the inner product with x_j for any fixed j:

$$0 = \left\langle \sum_{k=1}^{n}\alpha_k x_k, x_j\right\rangle = \sum_{k=1}^{n}\alpha_k\langle x_k, x_j\rangle = \alpha_j\|x_j\|^2.$$

Since $x_j \neq 0$, we have $\|x_j\|^2 > 0$, so $\alpha_j = 0$. Since j was arbitrary, all coefficients are zero. $\square$

Example 6.8 (Orthogonal Complement in $\mathbb{R}^3$)**.** Let $S = \{(1, 0, 0)\} \subseteq \mathbb{R}^3$. Then $S^{\perp} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 0\}$, which is the x_2x_3-plane. This is a closed subspace of $\mathbb{R}^3$ with $\dim(S^{\perp}) = 2$, and $\mathbb{R}^3 = \operatorname{span}(S) \oplus S^{\perp}$.

Example 6.9 (Orthogonal Complement in ℓ^2)**.** Let $S = \{e_1\} \subseteq \ell^2$, where $e_1 = (1, 0, 0, \ldots)$. Then

$$S^\perp = \{x \in \ell^2 : x_1 = 0\} = \{(0, x_2, x_3, \ldots) : \textstyle\sum_{i=2}^\infty |x_i|^2 < \infty\}.$$

This is a closed subspace of ℓ^2, and $\ell^2 = \operatorname{span}\{e_1\} \oplus S^\perp$.

Orthogonality in Value Function Approximation

In LSTD with features $\phi_1, \ldots, \phi_d$, the value function approximation $\hat{V} = \Phi w$ is the element of $M = \operatorname{span}\{\phi_1, \ldots, \phi_d\}$ closest to the Bellman target TV^* in the $L^2(\mu)$ norm. The optimality condition is precisely orthogonality: the residual $TV^* - \hat{V}$ must lie in $M^\perp$, i.e., $\langle TV^* - \hat{V}, \phi_j \rangle_\mu = 0$ for all j. This is the normal equation of linear regression.

7 Exercises

Exercise 7.1 (Verifying Inner Product Axioms)**.** For each of the following, verify that the given map is an inner product:

(a) $\langle x, y \rangle = x^\top A y$ on $\mathbb{R}^n$, where A is a symmetric positive definite matrix.
(b) $\langle f, g \rangle = \int_0^1 f(t)\, g(t)\, w(t)\, dt$ on $C[0, 1]$, where $w\colon [0, 1] \to (0, \infty)$ is continuous.
(c) $\langle f, g \rangle = f(0)g(0) + \int_0^1 f'(t)\, g'(t)\, dt$ on $C^1[0, 1]$.

Exercise 7.2 (Cauchy–Schwarz in Specific Spaces)**.** Write out the Cauchy–Schwarz inequality explicitly for:

(a) $\mathbb{R}^n$ with the standard inner product.
(b) ℓ^2 with the standard inner product.
(c) $L^2([a, b])$ with the standard inner product.

In each case, state the equality condition.

Exercise 7.3 (Parallelogram Law Fails for $p \neq 2$)**.** Show that the parallelogram law fails in $(\mathbb{R}^2, \|\cdot\|_p)$ for $p \neq 2$. Conclude that $\|\cdot\|_p$ does not arise from an inner product when $p \neq 2$.
Hint: Take $x = (1, 0)$ and $y = (0, 1)$ and compute both sides of the parallelogram law.

Exercise 7.4 (The Polarization Identity Determines the Inner Product)**.** Let $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ be two inner products on a vector space X that induce the same norm: $\langle x, x \rangle_1 = \langle x, x \rangle_2$ for all x. Use the polarization identity to show that $\langle x, y \rangle_1 = \langle x, y \rangle_2$ for all x, y.

Exercise 7.5 (Continuity of the Inner Product: Quantitative Bound)**.** Show that for all x, x', y, y' in an inner product space:

$$|\langle x, y \rangle - \langle x', y' \rangle| \leq \|x - x'\| \cdot \|y\| + \|x'\| \cdot \|y - y'\|.$$

Conclude that the inner product is Lipschitz on bounded sets.

Exercise 7.6 (Completeness of ℓ^2)**.** Give a detailed proof that ℓ^2 is complete (i.e., a Hilbert space):

(a) Let $(x^{(n)})$ be a Cauchy sequence in ℓ^2. Show that for each fixed i, the sequence $(x_i^{(n)})_{n \geq 1}$ is Cauchy in $\mathbb{R}$ (or $\mathbb{C}$) and define $x_i^* = \lim_n x_i^{(n)}$.
(b) Show that $x^* = (x_1^*, x_2^*, \ldots) \in \ell^2$.
(c) Show that $\|x^{(n)} - x^*\|_{\ell^2} \to 0$.

Exercise 7.7 (Orthogonal Complement of a Subspace of $\mathbb{R}^n$)**.** Let $M = \{x \in \mathbb{R}^4 : x_1 + x_2 = 0 \text{ and } x_3 - x_4 = 0\}$.

(a) Find a basis for M.
(b) Find a basis for $M^\perp$ (in the standard inner product).

(c) Verify that $\mathbb{R}^4 = M \oplus M^\perp$.

Exercise 7.8 (Pythagorean Theorem and Best Approximation)**.** Let M be a subspace of an inner product space X, $x \in X$, and $m \in M$. Show that if $x - m \perp M$ (i.e., $\langle x - m, y\rangle = 0$ for all $y \in M$), then m is the unique closest point to x in M:

$$\|x - m\| < \|x - m'\| \quad \text{for all } m' \in M,\ m' \neq m.$$

Hint: Write $x - m' = (x - m) + (m - m')$ and use the Pythagorean theorem.

Exercise 7.9 (Gram–Schmidt Orthogonalization)**.** Let x_1, x_2, x_3 be linearly independent vectors in an inner product space.

(a) Apply the Gram–Schmidt process: set $e_1 = x_1/\|x_1\|$, then $\tilde{e}_2 = x_2 - \langle x_2, e_1\rangle e_1$, then $e_2 = \tilde{e}_2/\|\tilde{e}_2\|$, and similarly for e_3. Show that $\{e_1, e_2, e_3\}$ is orthonormal and $\operatorname{span}\{e_1, \ldots, e_k\} = \operatorname{span}\{x_1, \ldots, x_k\}$ for each k.

(b) Apply this to $x_1 = (1, 1, 0)$, $x_2 = (1, 0, 1)$, $x_3 = (0, 1, 1)$ in $\mathbb{R}^3$ with the standard inner product.

Exercise 7.10 (The Fisher Information Inner Product)**.** Let $\pi_\theta(a|s)$ be a parameterized policy family with $\theta \in \mathbb{R}^d$. The Fisher information matrix is $F(\theta) = \mathbb{E}_{\pi_\theta}\left[\nabla \log \pi_\theta(a|s) \cdot \nabla \log \pi_\theta(a|s)^\top\right]$.

(a) Show that $F(\theta)$ is symmetric and positive semi-definite.

(b) Assuming $F(\theta)$ is positive definite, show that $\langle u, v\rangle_F = u^\top F(\theta)\, v$ defines an inner product on $\mathbb{R}^d$.

(c) The *natural gradient* is $\tilde{\nabla} J(\theta) = F(\theta)^{-1} \nabla J(\theta)$. Show that $\tilde{\nabla} J(\theta)$ is the steepest ascent direction of J in the Fisher metric, i.e., it solves $\max_{\|v\|_F = 1} \langle \nabla J(\theta), v\rangle$.

8 Summary of Key Results

Result	Statement (Informal)	RL/ML Relevance
Inner product (Def. 2.1)	Bilinear form with positivity, symmetry, and definiteness	Defines geometry on parameter and function spaces
Cauchy–Schwarz (Thm. 3.1)	$\lvert\langle x, y\rangle\rvert \leq \|x\| \cdot \|y\|$	Gradient bounds, cosine similarity
Triangle inequality (Cor. 3.2)	$\|x + y\| \leq \|x\| + \|y\|$	Norm is a valid distance measure
Parallelogram law (Thm. 4.1)	Characterizes norms from inner products	Identifies L^2 as the "right" space
Polarization (Prop. 4.4)	Inner product recovered from norm	Implicit inner product computation
IP continuity (Prop. 5.2)	$x_n \to x, y_n \to y \Rightarrow \langle x_n, y_n\rangle \to \langle x, y\rangle$	Stability of projections under approximation
Pythagorean (Thm. 6.2)	$x \perp y \Rightarrow \|x + y\|^2 = \|x\|^2 + \|y\|^2$	Bias-variance decomposition
Orth. complement (Prop. 6.5)	$S^\perp$ is a closed subspace	Normal equations in LSTD

Looking Ahead

With inner products, orthogonality, and Hilbert space structure in hand, we are ready for:

- **Lesson 20:** The Projection Theorem and Riesz Representation Theorem—the tools for orthogonal projection (least-squares approximation) and for identifying Hilbert spaces with their duals.
- **Lesson 21:** The fundamental theorems of functional analysis (Hahn–Banach, Uniform Boundedness, Open Mapping, Closed Graph)—powered by the Baire category theorem from Lesson 18.
- **Lesson 22:** Fixed point theorems (Banach, Brouwer, Schauder, Kakutani)—the convergence engine for value iteration and the existence proof for Nash equilibria.

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